

# Run-Level Verifiability for Sensor Fusion: Cumulative Guarantees and Capture-Resilience Profiles

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**Abstract.** Sensor-fusion deployments often retain only derived outputs, leaving an auditor unable to distinguish record alteration, credential misattribution, omission, and incorrect computation. Existing protocols prove different bundles of these properties under different compromise assumptions, which makes their claims and costs difficult to compare. We define four atomic run-level goals: externally anchored manifest integrity, credential attribution, policy-relative coverage, and conformance to the declared fusion computation. Their prefixes form a cumulative reporting hierarchy, while a separate capture-resilience profile states which software or hardware compromises the evidence tolerates. A cross-field map shows how secure logging, authenticated aggregation, succinct proofs, attestation, and robust fusion contribute to the goals without automatically constituting a complete level. We exercise the vocabulary with an executable model of nine complete stacks, one partial baseline, and five proof circuits for a cold-chain aggregation policy. The study yields three design rules: short authenticated records can cost less than commitments; sampling reduces communication only by accepting explicit false-acceptance risk; and credential attribution, rather than fusion arithmetic, dominates the demonstrated proof circuit. The artifact is a protocol model, not a device or physical-attack evaluation.

**Keywords:** Data fusion · Verifiability · Data provenance · Remote attestation · Resource-constrained devices.

## 1 Introduction

Distributed fusion systems reduce sensor streams to the quantities on which a decision is made. A pharmaceutical logger, for example, can reduce a temperature history to a mean kinetic temperature and an excursion decision [18]; a maritime system can reduce many reports to a track.

That reduction also removes the auditor’s simplest check: retain the inputs and recompute. An aggregator may alter a record, attach the wrong origin, omit an inconvenient observation, or report a value that the declared operator did not produce. Physical access can additionally expose application software and keys. Classical verifiable computation begins with a powerful worker and a weaker client that wants to avoid recomputation [10]. Here the evidence producer may

be the least capable device, although aggregation and proof generation can also be delegated.

General frameworks let an application choose a verified property, while concrete aggregation protocols prove fixed packages of properties. Proof-carrying data uses an application-defined compliance predicate [6]; the KTV framework represents verifiability through a goal, a judge, and an error bound [7]; and aggregation protocols combine authentication, commitments, and operator-specific checks [16,15,19]. They do not provide this particular cumulative, run-level vocabulary for saying which part of a package a deployment can verify.

We separate four atomic conditions—anchored manifest integrity, credential attribution, policy-relative coverage, and computation conformance—and use their prefixes as a cumulative reporting convention. The hierarchy is chosen because a claim about the policy-required computation should not omit the integrity, attribution, or coverage obligations on which that claim relies; it is not a claim that the atomic conditions have a unique logical order. Physical compromise is reported independently through a capture-resilience profile. No combination proves that a sensor measured the physical world honestly.

We contribute G1–G4 with cumulative V0–V4 and capture profiles; a cross-field mechanism map; and an executable cold-chain model of nine complete stacks, one partial baseline, and five proof circuits. The evaluation yields design rules about representation, probabilistic coverage, and attribution cost. The V prefix distinguishes this vocabulary from JDL fusion levels, which grade what is inferred rather than what can be checked [17].

Generative AI assisted literature discovery, artifact development, checks, and revision. Human authors verified all sources, code, analyses, and text and retain full responsibility.

## 2 Related Work

Definitions of verifiability separate goals from judges. Non-interactive verifiable computation addresses outsourcing [10], proof-carrying data distributes a compliance predicate [6], and Cortier et al. parameterize voting verifiability by a goal, judge, tolerance, and benign-run condition [7]. We instantiate that scaffold for one fusion run.

Verifiable aggregation protocols instead bundle assurances. SIA authenticates reports, commits to values, and checks operator-specific approximations with sampling and interactive proofs [16]. Hierarchical secure aggregation prevents intermediate manipulation beyond the values injected directly by compromised nodes [5]. Recent schemes add public verification for private summation [15] or designated verification of private inner products [19]. Their theorems are packages for particular operators and adversaries, not cumulative names for the run-level conditions inside those packages.

Mechanism surveys classify protocol families rather than claims. Bontekoe et al. [4] organize verifiable privacy-preserving computation by computation and verification technique; our map instead puts the atomic audit condition first.

Provenance and runtime assurance protect adjacent objects: SLSA orders build evidence [1], RATS appraises an attesting environment [2], and APEX proves execution under software compromise [9]. None is itself a cumulative V-level.

Robust fusion addresses what cryptography cannot validate. MATE infers which sensors to believe from cross-sensor consistency [12]; V-goals instead check bound records and computation. Privacy is likewise orthogonal unless its mechanism supplies an audit condition.

### 3 Proposal: Atomic Goals and Cumulative Levels

#### 3.1 System and Adversary Model

A fusion system consists of  $m$  sensors, an aggregator, and an off-system auditor. Before acquisition, an authenticated and versioned public policy  $P$  fixes the device roster, an epoch, the canonical sequence of eligible record identifiers, and the operator  $F$  with its public parameters. It also names the external anchoring procedure and verification material provisioned before the run. Policy authentication is a prerequisite: the goals below are relative to the particular  $P$  that the auditor accepts.

Sensors produce the policy-ordered sequence  $O = ((d_j, q_j, v_j))_{j=1}^n$ , where  $q_j$  is a device-local sequence number; the policy need not trust a device wall clock. A manifest  $M$  contains the ordered identifiers and binding commitments to their values. The aggregator publishes an output  $y$ , a manifest commitment  $c_M$ , and evidence  $\pi$ . Each run has a unique 16-byte identifier  $u$ . The auditor's view is  $\text{view}(r) = (P, u, y, c_M, \pi)$  plus the provisioned verification material. Its judge may also hold private protocol state  $\eta$ , including a fresh 32-byte anchor state  $a$  and a sampling challenge issued after observing an anchored  $c_M$ ;  $\eta$  is not accepted merely because the producer copied it into  $\pi$ .

The anchor signs  $H(P) \| u \| a \| c_M$ . Its trusted service guarantees fresh  $a$  and unique  $u$ , and the auditor records the anchor receipt before recording acceptance of  $y$ ; the signature alone is not a timestamp. For sampling, the auditor then issues a separate unpredictable challenge. A computationally bounded adversary may control the network and aggregator, modify stored records, forge an origin label, select which records reach  $M$ , or choose  $y$ . A capture profile separately states whether sensor application software or isolated secrets may also be compromised.

The fixed schedule covers closed windows. Event-triggered acquisition, churn, and open-world reports additionally require an authenticated closure rule, such as signed boundary markers; without one we do not claim policy-complete coverage. Order-sensitive filters consume values in  $P$ 's canonical order.

#### 3.2 Atomic Conditions and Reporting Levels

The four atomic conditions for a run are:

**Table 1.** Cumulative run-level reporting convention. Availability, physical truth, and suitability of  $F$  remain outside the hierarchy.

| Name                      | Prefix                 | What the name does not claim               |
|---------------------------|------------------------|--------------------------------------------|
| V0 None                   | no run-level condition | any evidentiary guarantee                  |
| V1 Anchored               | G1                     | availability or recoverability of openings |
| V2 Credential-attributed  | G1–G2                  | honest transduction or protected keys      |
| V3 Policy-complete        | G1–G3                  | events outside $P$ or open-window closure  |
| V4 Computation-conformant | G1–G4                  | physical truth or that $F$ is appropriate  |

- G1 (anchored integrity)** an external party bound  $u$ , fresh auditor state, and a binding  $c_M$  before accepting  $y$ , so replay or a later entry change is detected;
- G2 (credential attribution)** each value commitment in  $M$  is authenticated under the credential assigned by  $P$  to the device it names;
- G3 (policy-relative coverage)** the ordered identifier sequence in  $M$  equals the sequence required by  $P$ ; and
- G4 (computation conformance)**  $y = F(O_M)$  for policy-ordered openings  $O_M$  of all commitments in  $M$ .

These conditions are distinct dimensions. The V-levels are their cumulative prefixes:  $\gamma_0$  contains all runs and  $\gamma_i = \bigcap_{j=1}^i G_j$  for  $1 \leq i \leq 4$ . This prefix is a reporting convention chosen because a claim about the policy-required computation should not silently omit the integrity, attribution, or coverage obligations on which it relies. It does not imply that a mechanism contributing G2 or G4 must implement every earlier condition.

We instantiate the goal-judge-tolerance scaffold of Cortier et al. [7]. Let  $\mathcal{A}_\rho$  be the adversaries admitted by a fixed capture profile  $\rho$ .

**Definition 1 (Level- $i$  verifiability).** *A system  $\Sigma$  achieves  $\mathbf{Vi}$  under profile  $\rho$  with tolerance  $\delta$  if there is a probabilistic polynomial-time judge  $J$  such that, for every  $\mathcal{A} \in \mathcal{A}_\rho$ ,*

$$\Pr[r \leftarrow \Sigma^{\mathcal{A}} : r \notin \gamma_i \wedge J(\text{view}(r), \eta) = \text{accept}] \leq \delta,$$

*and  $J$  accepts a benign run in  $\gamma_i$  with overwhelming probability.*

Table 1 names the cumulative prefixes. The qualified names are intentional: a signature attributes a credential rather than proving an honest transducer reading, and coverage is complete only relative to  $P$ . By construction,  $\mathbf{Vi+1} \Rightarrow \mathbf{Vi}$  under the same profile. The adjacent counterexamples in Section 5.2 establish that the added conditions are non-redundant; they do not derive a unique total order from independence alone.

A concrete trace explains the cumulative choice. If  $P$  requires sequence numbers 1 through 60, a manifest containing authentic records 1 through 59 satisfies

**Table 2.** Capture-resilience templates reported independently of a V-level.

| Profile              | Adversary admitted                                                        | Evidence base required                                           |
|----------------------|---------------------------------------------------------------------------|------------------------------------------------------------------|
| $\rho_{\text{soft}}$ | network and aggregator control; no enrolled sensor application compromise | sensor credential in ordinary application memory                 |
| $\rho_{\text{iso}}$  | full application-software compromise                                      | isolated key, freshness, and measured acquisition/execution path |
| $\rho_{\text{eval}}$ | named physical attacks within an evaluation scope                         | laboratory-evaluated root of trust and key protection            |

G2 and fails G3. A proof that  $y = F(O_M)$  over that incomplete manifest could satisfy G4 in isolation but is not cumulative V4. Conversely, a homomorphic authenticator may contribute G2+G4 without constituting a complete V4 stack.

### 3.3 Boundaries and Exclusions

Mechanisms do not name levels. Defining V1 as “a Merkle tree,” for example, would make cost analysis circular and exclude equivalent constructions. The ALCOA+ vocabulary offers a regulatory reading, not a derivation: its Attributable, Complete, Enduring, and Accurate terms resemble parts of G1–G4, but legibility and availability are operational, consistency reaches across runs, and contemporaneity needs trusted time [14]. We stop at G4 because  $P$  already fixes  $F$  and its parameters. Software-build provenance is an adjacent object, as SLSA illustrates [1].

Capture also bounds what evidence can establish. A sensor controlling its own measurement path can commit and authenticate a plausible but false value, which later processing evidence cannot distinguish from an honest one [16]. Availability is separate too: a commitment proves binding only when an opening is supplied; it does not recover a discarded record.

### 3.4 Capture-Resilience Profiles

Physical capture changes which parties and secrets the judge may trust, not the run-level condition. Extracting a sensor’s signing key undermines G2 for that sensor, while compromising only the aggregator need not do so when sensors sign their own commitments. Every result must therefore pair its V-level with a capture profile.

Table 2 gives templates a deployment must refine. A  $\rho_{\text{eval}}$  claim must name the evaluated duration and attacks; hardware supports it only within that scope. RATS appraises evidence against references, endorsements, and policy [2]; APEX addresses software compromise, not general physical resistance [9].

No profile makes a physical observation true. A protected path may stop compromised software from changing a reading after acquisition, but cannot stop environmental spoofing or a compromised transducer from supplying a legal-

**Table 3.** Technique-to-goal map. A cumulative  $V_i$  claim requires every  $G1$ – $G_i$  condition, possibly by composition. “Direct” denotes the condition to which the technique principally contributes, not a universal theorem for the family.

| Field / technique                | Examples                                        | Direct                                            | Cumulative interpretation and caveat                                                                                                                |
|----------------------------------|-------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Tamper-evident logging           | Merkle/history trees; hash chains [8]           | G1 with an external witness                       | A producer-controlled root is not G1. A tree binds leaves and a chain binds an ordered prefix; neither proves coverage or availability of openings. |
| Record authentication            | MACs, signatures, BLS aggregation [3]           | G2                                                | Cumulative V2 also needs G1. It proves use of a credential; device identity depends on enrollment and honest acquisition on the capture profile.    |
| Secure sensor aggregation        | SIA [16]                                        | G1/G2 plus probabilistic, operator-specific tests | Its sampling theorem and approximate aggregate must be mapped to G3/G4 and $\delta$ ; approximate validity is not exact cumulative V4.              |
| Verifiable private aggregation   | mPVAS; PPVDA [15,19]                            | G2 and operator-specific G4                       | G1 and G3 are not automatic. Public versus designated verification, collusion, supported operator, and privacy assumptions remain explicit.         |
| General verifiable computation   | Groth16; proof-carrying data [11,6]             | G4 for the constrained statement                  | A circuit may internalize G2/G3, but freshness and external anchoring still supply G1. Setup, statement binding, and numerical semantics matter.    |
| Attestation / proof of execution | RATS; APEX; VECODI [2,9,13]                     | execution identity and G4 under an isolated path  | Cumulative use needs appraisal, enrollment, input binding, coverage, and a fresh external G1 event. Supports $\rho_{iso}$ , not physical truth.     |
| Roots of trust / evaluation      | Secure elements, TPMs, PUFs, evaluated hardware | no G-condition alone                              | Supports a capture profile only for the evaluated key, path, attacks, and duration; small report size is not a V-level.                             |
| Robust or statistical fusion     | MATE and trust-aware fusion [12]                | physical-value plausibility / resilience          | Orthogonal to G1–G4 and useful precisely because computation-conformant V4 cannot establish physical truth.                                         |

looking value. Cross-sensor trust estimation addresses this residual risk [12]; it composes with the evidence goals but is not implied by them.

## 4 Cross-Field Mechanism Map

Mechanisms attach to atomic conditions, not automatically to cumulative levels. Table 3 maps representative families by their direct contribution and the assumption needed to compose that contribution. A row marked G4 means conformance only to the statement or operator that the cited construction actually verifies; it does not silently supply G1–G3.

The map exposes several composition rules. Aggregate signatures compress tags, not messages, identities, key bindings, or coverage. A homomorphic MAC can authenticate a linear aggregate for a designated auditor while omitting the manifest integrity and per-record evidence needed by a cumulative stack. A succinct proof can instead constrain attribution and coverage inside its statement, but its manifest commitment still needs a fresh G1 anchor.

As a worked mapping, SIA’s authenticated query nonce and server-held Merkle commitment contribute G1; sensor–server MACs on opened leaves con-

tribute probabilistic G2; duplicate, count, and input-representation tests contribute probabilistic coverage under its ID-knowledge or alarm-channel assumptions; and its interactive tests establish operator-specific approximation [16]. It does not reach cumulative V3 here because G3 asks for the exact policy-ordered identifier sequence, nor V4 because G4 asks for exact  $y = F(O_M)$  rather than an  $\epsilon$ -approximation. Its own  $(\epsilon, \delta)$  claim remains valuable and should be reported directly.

Sampling adds a protocol condition rather than a new semantic goal. The root must be anchored before an unpredictable verifier challenge, and an offline judge must compare the response with its own stored challenge or an authenticated challenge transcript. Trusting a challenge copied from the producer’s proof permits root grinding. Public and designated verification, and deterministic and probabilistic guarantees, must therefore be stated even when they contribute to the same G-condition.

Proofs of execution occupy a different trust profile from cryptographic arguments. Groth16 work follows the circuit [11], whereas APEX binds an output to execution under a hardware root [9]. In our modeled attestation stack, an external authority first anchors the manifest root; the isolated path then reports that root and  $y$ . The row measures only this protocol transcript under  $\rho_{\text{iso}}$ : it does not measure hardware, energy, or physical resistance.

## 5 Executable Evaluation and Cost Analysis

The evaluation asks whether the judges realize the atomic distinctions, which representation choices determine communication, and which statement layers dominate a succinct proof. The goal is design insight from an executable model, not a claim of device feasibility.

### 5.1 Artifact and Method

The artifact implements nine complete stacks, one deliberately partial baseline, five circuits, and four adjacent-condition attacks. A complete stack carries evidence for every G-condition through its claimed level. The seeded generator, raw JSON, table generator, numeric claim checker, and adversarial proof-object tests are included in the anonymous supplement at <https://anonymous.4open.science/r/edge-fusion/>.

We report  $\text{cost}(S, n, m, F) = \langle \mathbf{N}_P, \mathbf{N}_J, B_{\text{ev}}, M_P, M_J \rangle$ . The first two coordinates count named evidence-generation and judge primitives.  $B_{\text{ev}}$  is the online transcript beyond the pre-provisioned policy  $P$  and claimed output  $y$ , including verifier challenges.  $M_P$  and  $M_J$  are protocol state retained between records under the declared streaming schedule, not process RSS. If  $B_P$  bytes of policy and setup material are distributed over  $R$  runs, the amortized communication is  $B_{\text{ev}} + B_P/R$ . The same rule applies to device keys, policy rosters, anchor keys, and proof verification keys. The generator populates a canonically serialized, versioned Ed25519/BLS roster;  $H(P)$  binds its exact enrolled keys.

The workload contains  $n = 60$  scheduled readings from  $m = 4$  loggers and tests arithmetic mean, excursion count, and bounded fixed-point mean kinetic temperature (MKT). It is synthetic scalar sensor-data aggregation, not an evaluation of tracking, association, heterogeneous-source fusion, or pharmaceutical prevalence. The stack sweep covers  $n \in \{12, 20, 60, 120, 240, 480, 960\}$ . Groth16 experiments use circom 2.2.3, circomlib 2.0.5, and snarkjs 0.7.5 on an Apple M4 Pro with 24 GB of memory.

## 5.2 Separation and the Price of Sampling

The adjacent traces alter a record after anchoring (A1), use a credential that does not match the named device (A2), omit a required identifier (A3-short), or change the output of a complete authentic run (A4). Replaying each trace against each stack’s own evidence gives 36 complete-stack decisions; 34 are deterministic and all follow the cumulative definition. These executions are counterexamples separating adjacent goals, not security proofs for all adversaries. Separate mutations replace an enrolled key or the accepted roster; both are rejected.

The sampled V3 stack adds a harder coverage trace, A3-padded: it keeps length  $n$  by replacing the omitted identifier with an authentic duplicate. The auditor anchors the root, stores its own unpredictable 32-byte challenge, and checks exactly the  $k$  derived policy positions, paths, signatures, and identifiers. The challenge is not read from the producer’s proof object. A3-short is rejected before sampling because its length violates  $P$ ; A3-padded is detected only if a defective position is opened.

If  $t$  of  $n$  positions are defective before the challenge and  $k$  distinct positions are sampled uniformly without replacement, then

$$\Pr[\text{miss}] = \delta = \frac{\binom{n-t}{k}}{\binom{n}{k}}, \quad \Pr[\text{detect}] = 1 - \delta,$$

with  $\binom{n-t}{k} = 0$  when  $k > n-t$ . For one defect, detection is  $k/n$ . Thus  $n = 60, k = 20$  checks one third of the manifest but still accepts a fixed single-position defect with probability  $2/3$ . As an implementation sanity check, 2,000 fixed challenges detected A1 at 0.338 and A3-padded at 0.317, consistent with  $1/3$ . The formula does not cover a producer that learns or grinds the challenge before committing.

**Design rule 1:** sampling is attractive only when the policy explicitly accepts sparse-defect false acceptance; it is not a strong defense against one strategic omission.

## 5.3 Representation, Not the V-Number, Drives Bandwidth

For the declared serialization, the dominant byte formulas are

$$\begin{aligned} B_{\text{raw}}(n) &= 144 + n(6 + 64), \\ B_{\text{Ed}}(n) &= 144 + n(36 + 64), \\ B_{\text{BLS}}(n) &= 240 + 36n, \\ B_{\text{sample}}(n, k) &= 176 + k'(100 + 32\lceil \log_2 n \rceil), \quad k' = \min(k, n). \end{aligned}$$



**Table 4.** Decision rules from the executable model. Byte comparisons use  $n = 60$  and exclude pre-provisioned  $P$  and claimed  $y$ . The attestation row is a cost model; all others are executed.

| Design choice                  | Finding                                                                     | Decision implication                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Raw records vs. commitments    | Raw cumulative V4 is 4 344 B; commitment-based V2 is 6 144 B.               | Recompute when records are shorter than commitments and disclosure is acceptable.                          |
| Exact vs. sampled coverage     | Exact V3 reuses the 6 144-B V2 manifest; $k = 20$ sampling uses 6 016 B.    | Sampling saves only at scale and accepts an explicit false-acceptance bound.                               |
| Ed25519 vs. BLS                | BLS reduces V2 to 2 400 B but uses 61 verifier pairings.                    | Choose from auditor work and public-verification needs, not tag size alone.                                |
| Succinct attribution           | In-circuit EdDSA expands the mean circuit $17.1\times$ .                    | Keep authentication outside the circuit or choose proof-friendly credentials when the trust model permits. |
| <i>Modeled:</i><br>attestation | A root, external anchor, and report total 208 B under $\rho_{\text{iso}}$ . | Compare the isolated-path and appraisal assumptions before comparing bytes.                                |

The constants expose the boundary: each anchored stack carries a 32-byte root, 64-byte signature, 16-byte run ID, and 32-byte auditor state; 36 bytes is an identifier and commitment. Sampled openings also carry a Merkle path. Exact V3 sends the same full manifest as Ed25519 V2, so it adds judge work but no bytes.

At  $n = 60$ , raw authenticated records and recomputation provide cumulative V4 in 4,344 bytes, while commitment-based Ed25519 V2 uses 6,144 bytes. Here a 6-byte observation is smaller than its 36-byte manifest entry. With fixed  $k = 20$ , the sampled stack uses 6,016 bytes at  $n = 60$ ; accounting for Merkle-depth steps, it first becomes smaller than raw transmission at  $n = 94$  (the first measured point past that crossover is  $n = 120$ ). It buys that reduction with interaction and  $\delta$ , not with a stronger primitive.

The setup comparison also separates encodings. At  $n = 12$ , snarkjs emits 983 bytes of proof and public-signal JSON; the G1 signature, run ID, and auditor state bring this implementation bundle to 1,095 bytes. Packed raw V4 uses 984 bytes, so this cross-encoding implementation comparison is 111 bytes larger. Under normalized compressed-binary encoding, the 128-byte proof, 96-byte public fields, and 112-byte G1 context total 336 bytes—648 bytes smaller than packed raw. The verification key is 3,290 bytes as JSON or 352 bytes normalized; a first run is therefore 4,385 bytes in JSON or 688 bytes normalized. The  $n = 60$  attributed circuit was compiled but not proven, so we make no comparison there.

Aggregate signatures expose a different trade. BLS reduces V2 evidence from 6,144 to 2,400 bytes but replaces 61 Ed25519 verifications with 61 pairings and one aggregate-signature verification. The 16-byte homomorphic MAC is smaller still, but authenticates only a linear aggregate for a designated auditor. It omits G1 and per-record G2 evidence and is therefore a partial baseline, not V4.

The modeled attestation stack is visually separated in Table 4. Its 208-byte transcript contains a 32-byte manifest root, 48-byte run context, a 64-byte external anchor that supplies G1 before output acceptance, and a 64-byte isolated-

**Table 5.** Constraint waterfall at  $n = 60$ . Operator rows are alternatives over the binding layer; attribution is added to the arithmetic mean. The attributed row was compiled but not proven at  $n = 60$  and still requires external G1.

| Layer                        | Constraints | Increment | Direct role      |
|------------------------------|-------------|-----------|------------------|
| input binding                | 32 040      | —         | manifest binding |
| + arithmetic mean            | 32 041      | +1        | G4 arithmetic    |
| + excursion count            | 33 301      | +1 261    | G4 comparison    |
| + bounded MKT                | 38 631      | +6 591    | G4 approximation |
| + EdDSA attribution and mean | 549 392     | +517 351  | G2–G4            |

path report binding  $P$ , the root, and  $y$ . It assumes enrollment, appraisal, and policy enforcement under  $\rho_{\text{iso}}$ ; no secure element or device was measured.

**Design rule 2:** compare complete evidence objects and their trust and setup boundaries. A higher V-number need not cost more than a lower one, and a small primitive is not automatically a complete stack.

#### 5.4 Attribution Dominates the Demonstrated Circuit

Table 5 decomposes the proof statement. Binding 60 policy-ordered values takes 32,040 constraints. Arithmetic mean adds one; excursion counting adds 1,261 (3.9%); and bounded MKT adds 6,591 (20.6%). The MKT circuit constrains the declared 0–35 °C band at scale  $2^{16}$ . Integer divisions and the logarithm round toward  $-\infty$  with constrained remainders; the degree-12 approximation differs from floating point by 1.8 mK on this trace. These are alternative G4 operators over the same binding layer.

Adding policy-positioned EdDSA attribution to the arithmetic mean raises the circuit from 32,041 to 549,392 constraints, a factor of 17.1. It constrains G2–G4 and exposes  $c_M$ , but an external event must still supply G1. The  $n = 60$  attributed circuit was compilation-only. At  $n = 12$ , where the available setup sufficed, proving consumed a median 2,467 MB of resident memory over three trials (range 1,812–2,494 MB). This host measurement is evidence against assuming logger feasibility, not a device benchmark.

**Design rule 3:** optimizing the fusion formula alone will not make this proof deployable while conventional in-circuit attribution dominates. Keeping authentication outside the circuit or using proof-friendly credentials may be more consequential, when the resulting trust model is acceptable.

#### 5.5 Threats to Validity

The artifact is an executable protocol model, not deployable security software. Fixture keys are publicly derivable, the Groth16 setup is for benchmarking, and the isolated attestation path is assumed. Primitive and serialized-byte counts are exact for the code, but retained state is analytical and prover RSS is host-specific. We report no device latency, energy, radio duty cycle, or physical-attack

result. Commitment openings also require retained source data; availability and audit-storage cost are outside the cost vector.

One synthetic seed and scalar reductions cannot establish general fusion workload realism. Order-sensitive tracking or filtering can use the canonical policy order, but its circuit and protocol costs remain unmeasured. The sampling expression assumes a defect set fixed before a sound challenge; 32-bit draws use rejection sampling for exact uniformity. adaptive corruption and open-window acquisition need additional protocols. Finally, MKT correctness is relative to one bounded numerical specification, not all temperature bands or implementations.

## 6 Conclusion

Run-level assurance is clearer when the claim is separated from the mechanism and from capture resilience. G1–G4 identify anchored manifest integrity, credential attribution, policy-relative coverage, and computation conformance; V1–V4 are their cumulative prefixes. The cross-field map shows why a signature, succinct proof, or attestation report contributes to particular conditions without automatically constituting a complete level.

The executable study yields three lessons. Short authenticated records can be cheaper than commitment entries. Sampling saves disclosure only under a stated false-acceptance probability. In the demonstrated proof, credential attribution dominates the scalar fusion arithmetic. These findings compare protocol evidence, not deployability on a physically exposed device.

For deployment: authenticate and version  $P$ ; choose the V-claim and tolerance; state the capture profile; cover every required G-condition; and account for online evidence, setup, retention, and judge work separately. Future work should test an order-sensitive multi-sensor operator, a real attested path under physical evaluation, and adaptive or dynamic-window adversaries.

## References

1. SLSA build track basics, version 1.2. <https://slsa.dev/spec/v1.2/build-track-basics> (2026), accessed 2026-08-23
2. Birkholz, H., Thaler, D., Richardson, M., Smith, N., Pan, W.: Remote ATtestation procedures (RATS) architecture. Tech. Rep. RFC 9334, Internet Engineering Task Force (2023). <https://doi.org/10.17487/RFC9334>
3. Boneh, D., Gentry, C., Lynn, B., Shacham, H.: Aggregate and verifiably encrypted signatures from bilinear maps. In: Advances in Cryptology – EUROCRYPT 2003. LNCS, vol. 2656, pp. 416–432 (2003). [https://doi.org/10.1007/3-540-39200-9\\_26](https://doi.org/10.1007/3-540-39200-9_26)
4. Bontekoe, T., Karastoyanova, D., Turkmen, F.: Verifiability for privacy-preserving computing on distributed data – a survey. International Journal of Information Security **24**(3), 141 (2025). <https://doi.org/10.1007/s10207-025-01047-7>, arXiv:2309.08248
5. Chan, H., Perrig, A., Song, D.: Secure hierarchical in-network aggregation in sensor networks. In: ACM Conference on Computer and Communications Security (CCS). pp. 278–287 (2006). <https://doi.org/10.1145/1180405.1180440>

6. Chiesa, A., Tromer, E.: Proof-carrying data and hearsay arguments from signature cards. In: *Innovations in Computer Science (ICS)*. pp. 310–331. Tsinghua University Press (2010)
7. Cortier, V., Galindo, D., Küsters, R., Müller, J., Truderung, T.: SoK: Verifiability notions for e-voting protocols. In: *IEEE Symposium on Security and Privacy (S&P)*. pp. 779–798 (2016). <https://doi.org/10.1109/SP.2016.52>
8. Crosby, S.A., Wallach, D.S.: Efficient data structures for tamper-evident logging. In: *USENIX Security Symposium*. pp. 317–334 (2009)
9. De Oliveira Nunes, I., Eldefrawy, K., Rattanaivanon, N., Tsudik, G.: APEX: A verified architecture for proofs of execution on remote devices under full software compromise. In: *USENIX Security Symposium*. pp. 771–788. USENIX Association (2020), <https://www.usenix.org/conference/usenixsecurity20/presentation/nunes>
10. Gennaro, R., Gentry, C., Parno, B.: Non-interactive verifiable computing: Outsourcing computation to untrusted workers. In: *Advances in Cryptology – CRYPTO 2010*. LNCS, vol. 6223, pp. 465–482. Springer (2010). [https://doi.org/10.1007/978-3-642-14623-7\\_25](https://doi.org/10.1007/978-3-642-14623-7_25), Cryptology ePrint Archive, Report 2009/547
11. Groth, J.: On the size of pairing-based non-interactive arguments. In: *Advances in Cryptology – EUROCRYPT 2016*. pp. 305–326 (2016). [https://doi.org/10.1007/978-3-662-49896-5\\_11](https://doi.org/10.1007/978-3-662-49896-5_11)
12. Hallyburton, R.S., Pajic, M.: Security-aware sensor fusion with MATE: the multi-agent trust estimator. In: *ACM Conference on Computer and Communications Security (CCS)* (2025). <https://doi.org/10.1145/3719027.3765193>, arXiv:2503.04954
13. Kiri, M.K., De Oliveira Nunes, I., Francillon, A., Rattanaivanon, N.: Verifiable and confidential DNN inference on low-end edge devices. arXiv preprint arXiv:2606.07470 (2026). <https://doi.org/10.48550/arXiv.2606.07470>
14. Medicines and Healthcare products Regulatory Agency: GxP data integrity guidance and definitions (2018), <https://www.gov.uk/government/publications/guidance-on-gxp-data-integrity>
15. Palazzo, M., Dekker, F.W., Brighente, A., Conti, M., Erkin, Z.: Privacy-preserving data aggregation with public verifiability against internal adversaries. In: *33rd USENIX Security Symposium*. pp. 6957–6974. USENIX Association (2024), <https://www.usenix.org/conference/usenixsecurity24/presentation/palazzo>
16. Przydatek, B., Song, D., Perrig, A.: SIA: Secure information aggregation in sensor networks. In: *ACM Conference on Embedded Networked Sensor Systems (SenSys)*. pp. 255–265 (2003). <https://doi.org/10.1145/958491.958521>
17. Steinberg, A.N., Bowman, C.L., White, F.E.: Revisions to the JDL data fusion model. In: *Sensor Fusion: Architectures, Algorithms, and Applications III*. Proc. SPIE (1999)
18. United States Pharmacopeia: General chapter 1079.2: Mean kinetic temperature in the evaluation of temperature excursions during storage and transportation of drug products. USP–NF (2023). [https://doi.org/10.31003/USPNF\\_M13855\\_02\\_01](https://doi.org/10.31003/USPNF_M13855_02_01)
19. Zhou, F., Wu, Q., Wu, P., Xu, J., Feng, D.: Privacy-preserving and verifiable data aggregation for internet of vehicles. *Computer Communications* **218**, 198–208 (2024). <https://doi.org/10.1016/j.comcom.2024.02.022>